# A/B Testing Analysis to Improve Marketing Campaign Conversion Rates

**Product Analyst Project**

Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy) | Jupyter Notebook

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# Business Problem

## Context

Company runs two marketing campaigns: general PSA and dedicated 'ad' campaign. Goal is driving user conversions through purchases.

## Problem

Unclear whether dedicated 'ad' campaign significantly outperforms standard PSA campaign. Marketing budget may be misallocated.

## Why It Matters

Understanding conversion rate differences is crucial for optimizing marketing spend, maximizing ROI, and improving revenue.

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# Hypothesis

## Null Hypothesis ($H_0$)

There is **no difference** in conversion rate between 'ad' and 'psa' campaigns.

Conversion_Rate_ad = Conversion_Rate_psa

## Alternative Hypothesis ($H_1$)

There **is a difference** in conversion rate. The 'ad' campaign leads to higher conversion.

Conversion_Rate_ad > Conversion_Rate_psa

### Primary Metric

Conversion Rate (Percentage of users who converted)

### Secondary Metric

Number of ads seen per user (to gauge engagement)

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# Experiment Design

**Goal:** Determine causal impact of 'ad' campaign on user conversion through randomized controlled experiment.

### Control Group

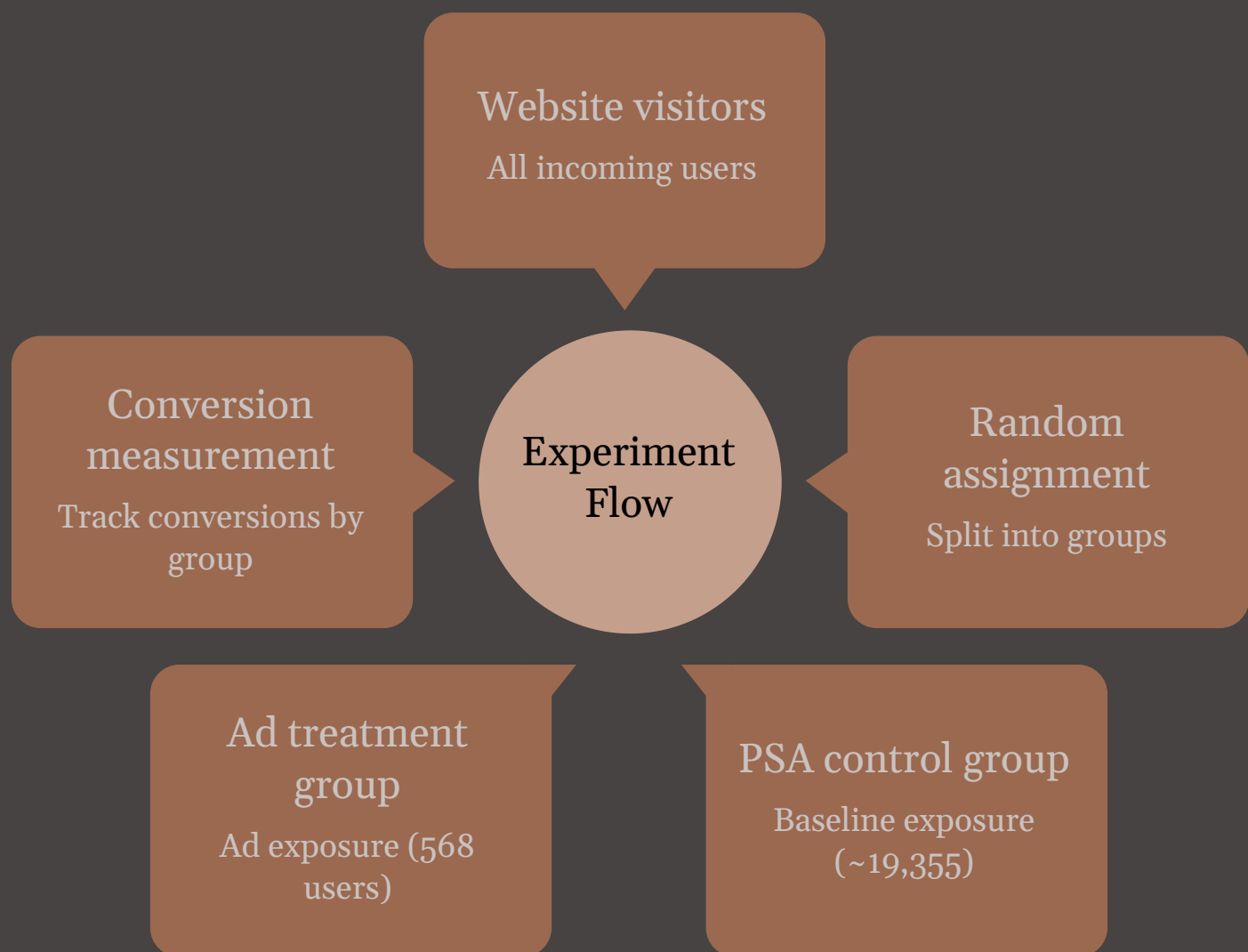
Users exposed to PSA campaign (baseline)

19,355 users (~3.3%)

### Treatment Group

Users exposed to ad campaign

568,746 users (~96.7%)

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# Dataset Overview

The dataset *marketing_AB.csv* captures user interactions with marketing campaigns.

| Column Name | Description | Data Type |
|---|---|---|
| user id | Unique identifier for each user | Integer |
| test group | Which campaign user saw: ad or psa | Categorical |
| converted | Whether user made purchase (True/False) | Boolean |
| total ads | Total number of ads user saw | Integer |
| most ads day | Day of week user saw most ads | Categorical |
| most ads hour | Hour of day user saw most ads | Integer |

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# Data Cleaning & Preparation

01

## Duplicate Check

Verified no duplicate entries based on user id, ensuring each user counted once

02

## Column Removal

Dropped unnamed index column and user id post-duplicate check as not needed for core analysis

03

## Variable Verification

Confirmed categorical variables had expected number of unique levels

04

## Class Imbalance Note

Observed significant imbalance: treatment group much larger than control group

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# Exploratory Data Analysis

Initial exploration revealed key insights into user behavior and conversion patterns.

### Overall Conversion

~97.5% did not convert, typical for e-commerce marketing environment

### Day Patterns

Mondays and Tuesdays higher; Saturdays lowest conversion rates

### Hour Patterns

Afternoon (15:00-17:00) and early evening (20:00) highest; late night lowest

### Ad Engagement

Median 13 ads seen; converted users saw higher median number of ads

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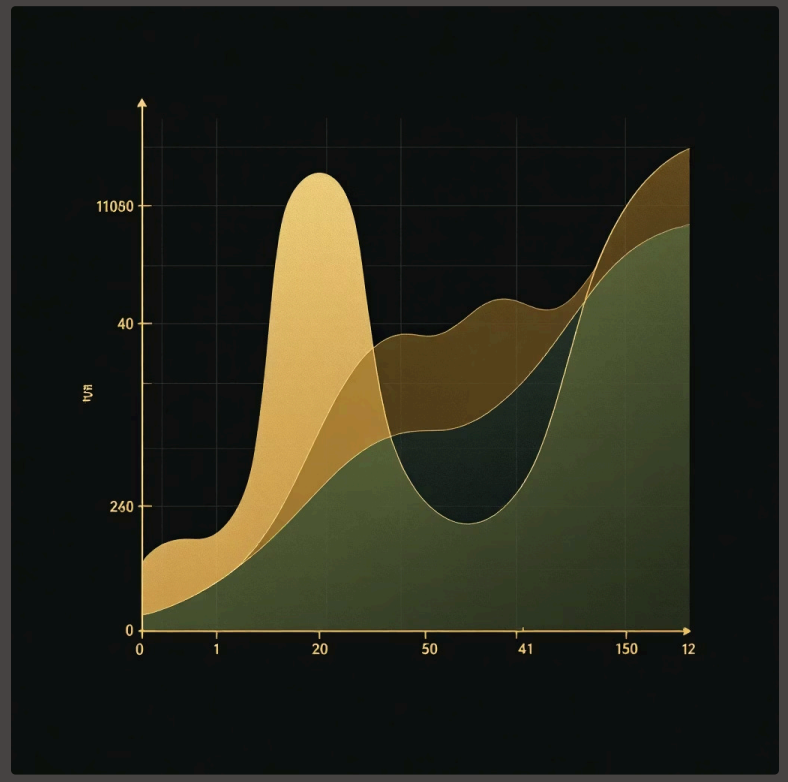
# Statistical Test

## Chi-Square Test of Independence

Used to determine if conversion rate difference between 'ad' and 'psa' groups is statistically significant.

**Method:** Contingency table compared observed vs. expected conversion frequencies. Chi-Square test calculates probability (p-value) that observed difference is due to random chance.

**Significance Level (α):** 0.05 (standard threshold in product analytics)

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# Results

Strong evidence that 'ad' campaign outperforms 'psa' campaign.

| Metric | Control Group (psa) | Treatment Group (ad) |
|---|---|---|
| Sample Size | 19,355 users | 568,746 users |
| Converted Users | 342 users | 14,517 users |
| **Conversion Rate** | **1.77%** | **2.55%** |
| **Relative Uplift** | - | **+44.1%** |

54.01

Chi-Square Statistic

1.99e-13

P-value

Far below 0.05 threshold

**Interpretation:** P-value is 0.000000000000199, much lower than significance level of 0.05. We **reject the null hypothesis**. The difference in conversion rates is **statistically significant**. The 'ad' campaign leads to a meaningfully higher conversion rate.

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# Product Insights

Why did the 'ad' campaign perform better?

### Relevance & Intent

'Ad' campaign likely contained product-specific messaging that resonated with users in shopping mindset. Generic 'psa' may not have provided same motivation or call-to-action.

### Engagement

Users who converted generally saw more ads. 'Ad' campaign more effective at driving repeated engagement leading to conversion over time.

### Behavioral Timing

Higher conversion rates during afternoons and early evenings suggest prime time to capture attention, leveraged by more effective 'ad' campaign.

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# Thank You!

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